

Adding Time to Deep Neural Networks

Standard Dynamical System

$$\begin{aligned}s^{(t)} &= f(s^{(t-1)}; \theta) \\ s^{(t)} &= f(s^{(t-1)}, x^{(t)}; \theta) \quad \text{External signal } x^{(t)}\end{aligned}$$

Recurrent Hidden Layer for State

$$h^{(t)} = f(h^{(t-1)}, x^{(t)}; \theta)$$

Unfolding Computation Graphs

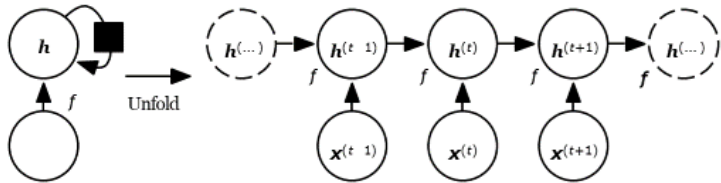
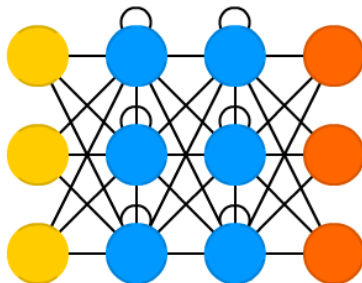


Figure 10.2

Adding Time to Deep Neural Networks

- Some nodes are allowed to link to previous versions of themselves

Recurrent Neural Network (RNN)

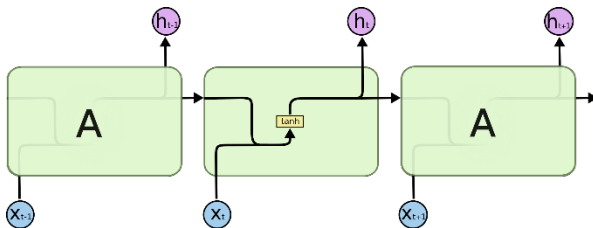


Problems with RNNs

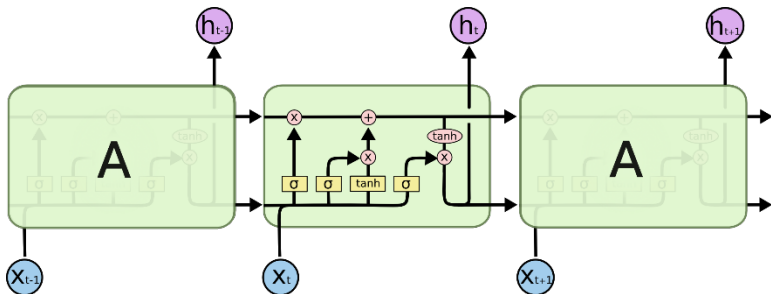
- Vanishing Gradients
- Long term dependencies
 - How far back to remember?
 - “I grew up in *China*. I speak fluent *Chinese*.”
 - If gap is short, RNN might be fine.
 - What is gap is very long?
- In theory any dependency can be modelled in an RNN
 - But in practice could be very hard to learn if you don't know how far back it was.

LSTM: Long-Term Short-Term networks

- More complex form of RNN
 - Doing long term dependencies the “right” way
- Integrate strategically remembering particular information from the past
- Formalizes a process for forgetting information over time.



<http://colah.github.io/posts/2015-08-Understanding-LSTMs/>



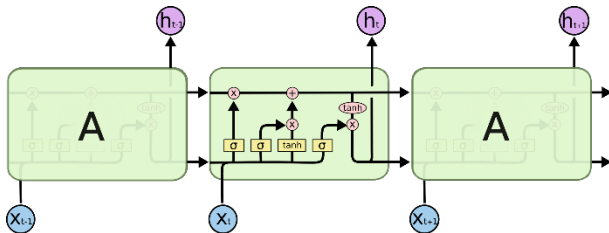
Neural Network
Layer

Pointwise
Operation

Vector
Transfer

Concatenate

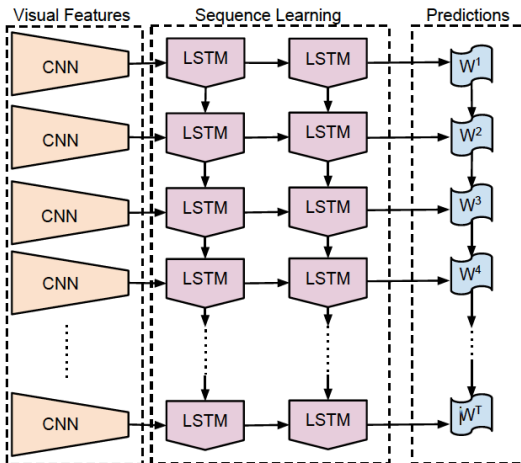
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- Process new input, use remembered variables
- Decide to keep or throw away some memory
 - Eg. Keep track of “country” until new person starts talking
- Decide on output data to send along to next step
- All weights for these decisions trained by back propagation

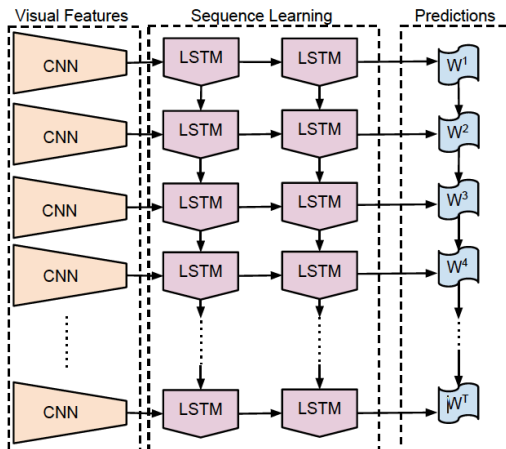
Long-term Recurrent Convolutional Networks (LRCN)

- Combine the abilities of CNNs and LSTMs into an end-to-end architecture



Applications of LRCNs

- **Activity Recognition** - Assign a class label for a video



Applications of LRCNs

- **Video prediction** - Can learn a generative model which can directly generate a new instance of a later timestep
 - Like an autoencoder on entire images between video frames

